EXECUTIVE BRIEFING | HEALTHCARE DECISION SUPPORT

Understanding Diabetes Progression

Baseline risk indicators, predictive performance, and clinical boundaries across 442 patients.

Diogenis Xenidis — LEAP Data Analysis & Statistics

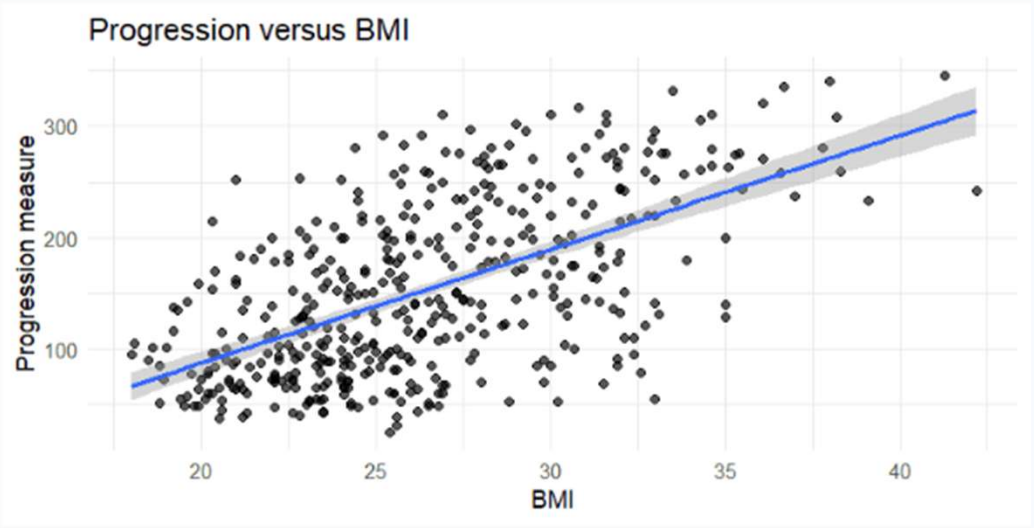
01 / COHORT & PRIMARY DRIVER

Higher BMI Tracks With Faster Progression

442 Complete Records: Zero missing values across ten baseline biomarkers.

Linear Trajectory: Disease progression rises steadily with baseline BMI.

GDPR Compliance: Anonymized records require strict clinical governance.



02 / BIOMARKER SELECTION

Three Independent Markers Drive Prediction



Body Mass Index

Strongest individual progression predictor.



Triglycerides & BP

Vital non-overlapping clinical signals.



Cholesterol Overlap

90% correlation creates redundancy.

Combining Indicators Captures 48% of Variation



Each +1 BMI unit adds +6.5 progression points (scale 25–346) — modest clinical shift per single unit.

What the Evidence Supports vs. What to Avoid

Supported by Analysis

Early Risk Screening: Flag patients with elevated BMI, blood pressure, and triglycerides.

Streamlined Panels: Drop redundant cholesterol markers from combined scoring.

52.3% Unexplained Risk: Diet, exercise, and medication adherence remain unmeasured.

Do Not Conclude

No Proven Causation: Observational correlations cannot guarantee direct intervention outcomes.

Not Automatic Reversal: Lowering BMI alone will not mechanically halt progression.

Lower Severe Precision: Predictions widen in error for extreme progression cases.

STRATEGIC RECOMMENDATION

Use BMI, Triglycerides, and Blood Pressure for Triage — Track Lifestyle Data Next

Deploy baseline models to prioritize patient monitoring while expanding clinical data collection under GDPR safeguards.